Curriculum Vitae for Ali Furkan Kalay

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+61 434 232 612 | alifurkan.kalay@mq.edu.au | alfurka.github.io

I am a Research Fellow at the Macquarie University Centre for Health Economy. I have a keen interest in leveraging novel large datasets, from social media to maps and administrative records, to extract meaningful insights, and in utilizing and advancing state-of-the-art statistical models.

PRIMARY: Health Economics, Applied Econometrics
SECONDARY: Statistics

Education

2020-2024 PH.D. IN ECONOMICS, The University of Queensland
2019 MASTER OF ADVANCED ECONOMICS, The University of Queensland
2016-2018 MASTER in Economics, Istanbul Technical University
2012-2016 BACHELOR OF SCIENCE in Industrial Engineering, Istanbul Technical University

Employment & Experience

2024-present *Research Fellow*, Macquarie University Centre for Health Economy
2019 - 2024 *Research Assistant & Tutor*, The University of Queensland
2017-2018 *Research Assistant*, Istanbul Technical University
2015 *Internship*, Product Development Department, Kuveyt Turk Participation Bank

Publications

2026 "LOST IN AGGREGATION: QUANTIFYING MEASUREMENT ERROR FROM GEOGRAPHIC CENTROIDS", *Economic Record*, (with Henry Wen). https://doi.org/10.1111/1475-4932.70072
2025 "GENERATING SYNTHETIC DATA WITH LOCALLY ESTIMATED DISTRIBUTIONS FOR DISCLOSURE CONTROL", *Australian & New Zealand Journal of Statistics*, 1–24. https://doi.org/10.1111/anzs.70032 [Software].
2019 "SPATIAL ANALYSIS OF TWITTER SENTIMENT AND DISTRICT-LEVEL HOUSING PRICES", *Journal of European Real Estate Research*, Vol. 12 No. 2, pp. 173-189, (with C. Hannum and K.Y. Arslanli)

Working Papers

2026 "WHEN BETTER AI WIDENS THE ADOPTION GAP IN HEALTH CARE" (with Jonas Fooken and Metin Uyanik)
2026 "SO IT BEGINS: HAS AI REACHED MATURE SOFTWARE?" (evidence from the Linux kernel and a 91-repository panel)
2026 "SUICIDE RISK AND PLACE: A SCOPING REVIEW OF REGIONAL MODELLING WITH ADMINISTRATIVE DATA ON SOCIO-STRUCTURAL DETERMINANTS" (with Anam Bilgrami, Meimei Chen, and Henry Cutler) (Revise and Resubmit, Social Science & Medicine)
2025 "BRIDGING LANGUAGE BARRIERS: THE IMPACT OF LARGE LANGUAGE MODELS ON ACADEMIC WRITING" (with Burak Dalaman and Nathan Kettlewell) (Reject and Resubmit, Journal of Information Technology, SSRN)
2024 "SOCIAL INTERACTIONS IN THE DEMAND FOR PRIVATE HEALTH INSURANCE: EVIDENCE FROM LINKED EMPLOYER-EMPLOYEE DATA" (with Alicia Rambaldi and Chris Rose)
2024 "LINKED ADMINISTRATIVE DATA AND ESTIMATING THE MARGINAL WILLINGNESS TO PAY FOR AMENITIES" (with James Hansen and Alicia N. Rambaldi) (Draft)
2023 "FORECASTING REGIONAL HOUSE SALES IN TURKEY USING GOOGLE TRENDS" (with Yi Zhang)(Draft)
2021 "DOUBLE ROBUST MASS-IMPUTATION WITH MATCHING ESTIMATORS" (arXiv)

Grants

2026 CI: "MQBS SEED FUNDING", *Macquarie Business School*, **AUD 9,701**.
2025-2029 CI: "UNDERSTANDING THE BROADER IMPACTS OF DISABILITY THROUGH THE AUSTRALIAN LINKED-ADMINISTRATIVE DATA", *Summer Foundation and Macquarie University.*

2024-2025 CI: "Predicting self-harm and suicide risk within Australian local government areas", *The National Suicide Prevention Office*, **AUD 296,196** (with Henry Cutler, Alan Woodward, and Anam Bilgrami)

2025 AI: "Spillovers Between Firm Performance and Employees' Health", *Business Research Grant Scheme, University of Technology Sydney*, **AUD 12,000** (with Nathan Kettlewell and Ali Vergili)

2025 CI: "Macquarie University ECR Enabling Scheme", **AUD 3,000**.

Research Supervision

PhD Kadek Hendra Darmawan

RA Henry Wen (2024), Hengzhe Zhao (2025-2026), Kadek Hendra Darmawan (2026)

Work-in-progress

"Forecasting Regional Suicide Risk in Australia Using Linked Administrative Data" (with Henry Cutler, Alan Woodward, and Anam Bilgrami)

"Spillovers Between Firm Performance and Employees" (with Nathan Kettlewell and Ali Vergili)

"Network Spillovers versus Common Factors: A Joint-Estimation Approach" (with James Morley, Valentyn Panchenko, Jorge M. Pinto, Chris Rose)

2025 "Baseline Health Distribution Using HILDA" (with Shan Jiang, Bonny Parkinson, and Yuanyuan Gu)

Talks & Lectures

2026 Conference: HTAi 2026 Annual Meeting (Istanbul, Türkiye), "Inequality in Quality-Adjusted Life Expectancy in Australia"

Seminar: Department of Economics, University of Auckland, "Prescriber Influence on the Adoption of New Medicines: Evidence from Movers in Australia"

Seminar: Department of Economics, University of Technology Sydney, "Prescriber Influence on the Adoption of New Medicines: Evidence from Movers in Australia"

Seminar: Internal Seminar (Macquarie University Centre for Health Economy), "Forecasting Regional Suicide Risk in Australia Using Linked Administrative Data"

Poster: National Suicide Prevention Conference 2026, "Forecasting Regional Suicide Risk in Australia Using Linked Administrative Data"

Workshop: 15th Australasian Workshop on Econometrics and Health Economics

Seminar: Centre for the Business and Economics of Health (CBEH), The University of Queensland

Workshop: Lifespan Health and Wellbeing Research Centre Workshop (Macquarie University), "The impact of depression and treatment uptake on labour productivity"

2025 Conference: Inaugural AE² Australasian Econometric Conference (Macquarie University City Campus, Sydney)

Seminar: Internal Seminar (Macquarie University Centre for Health Economy)

Conference: 46th Annual AHES Conference (Canberra, Australia)

Conference: International Health Economics Association (IHEA) Conference 2025 (Indonesia)

Conference: AI, Law and Society Conference (Macquarie University)

2024 Conference: 45th Annual AHES Conference

Conference: Internal Seminar (Monash University Centre for Health Economics)

Seminar: Internal Seminars (Macquarie University)

Presentation: on "Double Robust Difference-in-Difference Estimators" (Macquarie University Centre for Health Economy)

Conference: Australasian Workshop on Econometrics and Health Economics 2024

Seminar: Internal Seminar (Macquarie University)

Conference: Workshop on Economics of Health and Wellbeing (Macquarie University)

2023 Conference: 37^{th} PhD Conference in Economics & Business (University of Melbourne)

Conference: Econometric Society Australasian Meeting (University of New South Wales)

Conference: Australian Conference of Economists (Brisbane, Australia

Conference: Western Economic Association International (University of Melbourne)

Guest Lecturer: *Statistical Theory for Economists* (The University of Queensland)

Seminar: Brown Bag (The University of Queensland)

2022 Conference: Center for Efficiency and Productivity Analysis (The University of Queensland)
Presentation: *on Creating Synthetic Data* (Australian Bureau of Statistics)
2021 Presentation: *"Double Robust Mass-Imputation with Matching Estimators"* (Australian Bureau of Statistics)
Conference: 34^{th} PhD Conference in Economics & Business (Virtual)
2020 Seminar: PhD Colloquium (The University of Queensland)

Honors & awards

2024 Dean's Award for Outstanding HDR Thesis, School of Economics, The University of Queensland (Top 10% of HDR theses)
2020-2023 UQ Graduate School Scholarship, The University of Queensland (Stipend and tuition)
2019 Master of Advanced Economics Scholarship, The University of Queensland (Stipend and tuition)
2012-2016 EKOTEK: The Foundation of Economic and Technical Solidarity Success Grant (Monthly stipend)
2012-2016 KYK: State Higher Education Grant for Success (Monthly stipend)
2012 National University Entrance Exam (OSYM): ranked within 0.12% *(out of 2 million)*.
2011 Bronze Medal at Akdeniz University Math Olympics

Languages and Computer Skills

Languages **Turkish** (Native), **English** (Fluent), **Arabic** (Beginner)

Programming **Python** (Advanced, *synloc*), **R** (Advanced), **Stata** (Advanced), **Julia** (Intermediate) **Matlab** (Intermediate), **SQL** (Intermediate), **PHP** (Intermediate)